Texas A&M University
Department of Electrical and Computer Engineering

ECEN 474/704 – (Analog) VLSI Circuit Design

Spring 2018

Exam #1

Instructor: Sam Palermo

- Please write your name in the space provided below
- Please verify that there are **6 pages** in your exam
- You may use one double-sided page of notes and equations for the exam
- Good Luck!

Problem	Score	Max Score
1		40
2		40
3		20
Total		**100**

Name: _____

UIN: _____

Problem 1 (40 points)

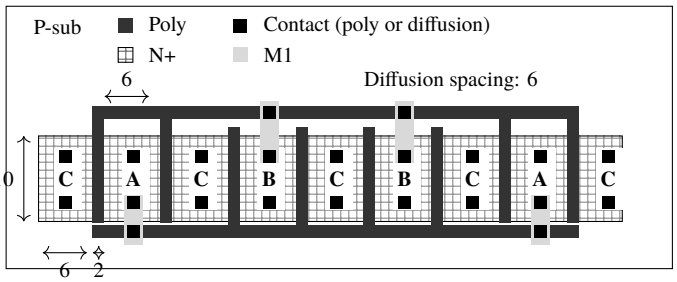
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_A = 2\,\text{V}$, $V_B = 1.5\,\text{V}$, $V_C = 1.0\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. $C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}$). The dimensions are given in μm, with the poly gates having $L_D = 0.1\,\mu$m.

a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors.

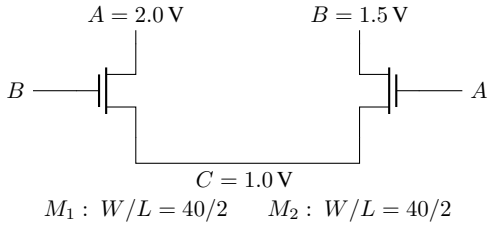
b) What region(s) are the transistors operating in?

c) **For node B only,** give an expression and calculate the total gate cap (only consider where B is the gate).

d) **For node A only,** give an expression and calculate the total junction cap, including the relevant channel-bulk cap. Note for the perimeter terms, include the sides underneath the gate.



Typeset solution. All eight individual transistors have $W/L = 10/2$. Four parallel devices form each equivalent transistor:



$M_1:\quad V_{GS} = 1.5 - 1.0 = 0.5\,\text{V} < V_T \quad \Longrightarrow$ cutoff or subthreshold,

$M_2:\quad V_{GS} = 2.0 - 1.0 = 1.0\,\text{V} > V_T, \quad V_{DS} = 0.5\,\text{V} > V_{GS} - V_T = 0.3\,\text{V} \quad \Longrightarrow$ saturation.

For B, M_1 is cut off. With $L_{\text{eff}} = 2 - 2(0.1) = 1.8\,\mu$m,

$$C_{gB} = C_{gc} + C_{gs,ov} + C_{gd,ov} = (40\,\mu\text{m})(1.8\,\mu\text{m})\frac{C_{ox}C_{jc}}{C_{ox}+C_{jc}} + 2(40\,\mu\text{m})C_{ov},$$

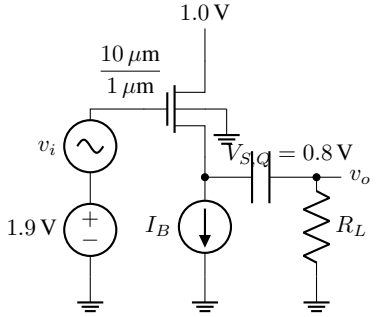
$$\boxed{C_{gB} = 72\,\mu\text{m}^2\frac{C_{ox}C_{jc}}{C_{ox}+C_{jc}} + 80\,\mu\text{m}\,C_{ov}.}$$

For A, the two 10×6 diffusion regions give

$$C_{jA} = A_A C_j + P_A C_{jsw} = 2(10)(6)\,\mu\text{m}^2 C_j + 2\,[2(10+6)]\,\mu\text{m}\,C_{jsw}, \qquad \boxed{C_{jA} = 120\,\mu\text{m}^2 C_j + 64\,\mu\text{m}\,C_{jsw}.}$$

Problem 2 (40 points)

For the circuit shown below, assume the transistor has $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, and $2\Phi_F = 0.9\,\text{V}$, and its DC drain current is determined by the following equations.



$$I_{DS} = I_o \exp\left(\frac{(V_{GS} - V_{Tn})q}{\zeta kT}\right)\left[1 - \exp\left(-\frac{V_{DS}q}{kT}\right)\right] \quad \text{(Subthreshold)},$$

$$I_{DS} = \mu_n C_{ox}\frac{W}{L - 2L_D}\left(V_{GS} - V_{Tn} - 0.5V_{DS}\right)V_{DS} \quad \text{(Triode)},$$

$$I_{DS} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L - 2L_D}\left(V_{GS} - V_{Tn}\right)^2\left(1 + \lambda V_{DS}\right) \quad \text{(Saturation)},$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right).$$

a. Calculate V_T and state the transistor's operation region.

$$V_T = 0.7 + 0.45\left(\sqrt{0.9 + 0.8} - \sqrt{0.9}\right) = \boxed{0.860\,\text{V}},$$

$$V_{DS} = 0.2\,\text{V}, \qquad V_{GS} - V_T = 1.1 - 0.86 = 0.24\,\text{V},$$

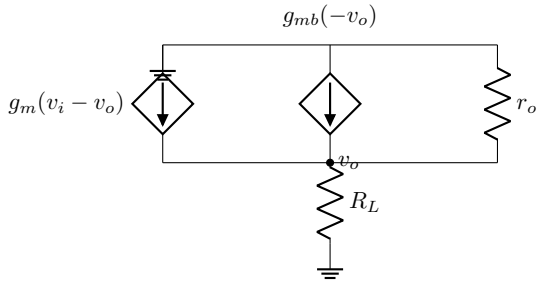
$$V_{DS} < V_{GS} - V_T \implies \boxed{\text{Operation region: Triode.}}$$

b. Using the appropriate I_{DS} equation above, **draw the low-frequency small-signal model of the circuit** (neglecting transistor capacitors), **include expressions for g_m, g_o and g_{mb}, and give an expression for the small-signal voltage gain** $A_v = v_o/v_i$ in terms of the relevant transistor conductances and load resistor. Assume the output AC-coupling capacitor acts as an ideal short.

$$g_m = \left.\frac{\partial I_{DS}}{\partial V_{GS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}V_{DS},$$

$$g_o = \left.\frac{\partial I_{DS}}{\partial V_{DS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}(V_{GS} - V_T - V_{DS}),$$

$$g_{mb} = \left.\frac{\partial I_{DS}}{\partial V_{BS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}V_{DS}\frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} = \frac{\gamma g_m}{2\sqrt{|2\Phi_F| + V_{SB}}} = \eta g_m.$$



$$v_o = \left[g_m(v_i - v_o) + g_{mb}(-v_o)\right](R_L \parallel r_o),$$

$$v_o\left[1 + (g_m + g_{mb})(R_L \parallel r_o)\right] = v_i g_m(R_L \parallel r_o),$$

$$\boxed{\frac{v_o}{v_i} = \frac{g_m(R_L \parallel r_o)}{1 + (g_m + g_{mb})(R_L \parallel r_o)}}.$$

c. In addition to the parameters given previously, assume $I_o = 1\,\text{nA}$, $kT/q = 25.9\,\text{mV}$, $\zeta = 1$, $\mu_n C_{ox} = 130\,\mu\text{A/V}^2$, $L_D = 0.1\,\mu\text{m}$, and $\lambda = 0.1\,\text{V}^{-1}$. Calculate the necessary load resistor for $|A_v| = 0.7\,\text{V/V}$.

$$A_v = \frac{g_m(R_L \parallel r_o)}{1 + (g_m + g_{mb})(R_L \parallel r_o)} = \frac{g_m}{G_L + g_o + (1+\eta)g_m},$$

$$G_L = \frac{g_m}{A_v} - g_o - (1+\eta)g_m,$$

$$g_m = (130\,\mu)\left(\frac{10}{0.8}\right)(0.2) = 325\,\mu\text{A/V},$$

$$g_o = (130\,\mu)\left(\frac{10}{0.8}\right)(1.1 - 0.86 - 0.2) = 65\,\mu\text{A/V},$$

$$\eta = \frac{0.45}{2\sqrt{0.9 + 0.8}} \simeq 0.173,$$

$$G_L = \frac{325\,\mu}{0.7} - 65\,\mu - 1.173(325\,\mu) \simeq 18.1\,\mu\text{A/V},$$

$$R_L = \frac{1}{G_L} \simeq \boxed{55.4\,\text{k}\Omega}.$$

Rounding note: the supplied solution rounds V_T, g_o, and η before computing G_L. Retaining their full values gives $R_L \simeq 55.83\,\text{k}\Omega$; the displayed $55.4\,\text{k}\Omega$ reproduces the supplied rounded calculation.

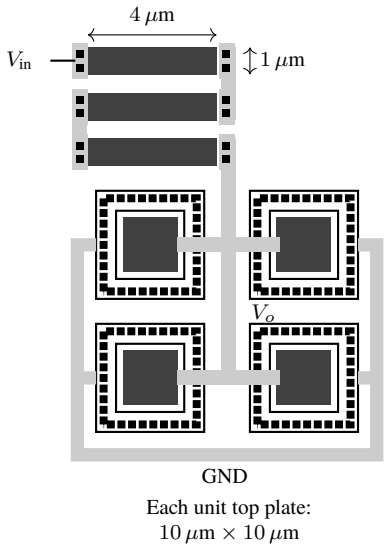
R_L for $|A_v| = 0.7\,\text{V/V} = 55.4\,\text{k}\Omega$

Problem 3 (20 points)

The layout of a simple RC-filter is shown below. For the 3-finger resistor, assume $R_\square = 200\,\Omega/\square$ and each contact has a resistance of $10\,\Omega$. The capacitor consists of 4 unit capacitors that each have top-plate dimensions of $10\,\mu$m $\times$ $10\,\mu$m. These unit capacitors have a capacitive density of $1\,\text{fF}/\mu\text{m}^2$.

a. Sketch the equivalent circuit and give the values of the resistor and capacitor. The resistor contacts should be included in its resistance calculation, but the capacitor contacts' resistance may be ignored. Neglect the impedance of all metal interconnect between the elements.

b. Sketch the Bode plot (magnitude only), and label any pole and/or zero frequencies.



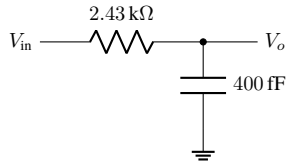
Typeset solution.

n is the number of parallel contacts at each end; F is the number of resistor fingers.

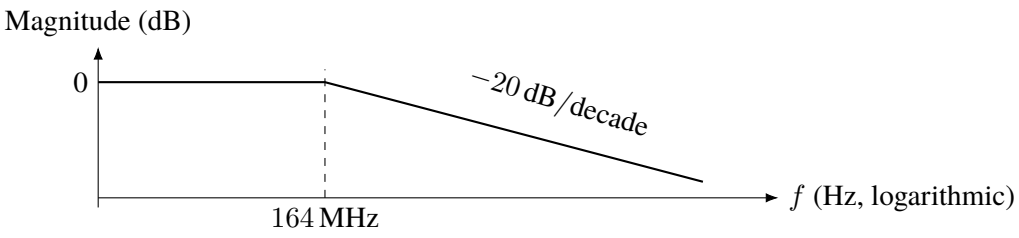
$$\begin{aligned} R_{\text{total}} &= \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F \\ &= \left[\frac{2(10)}{2} + 200\left(\frac{4}{1}\right)\right] 3 \\ &= \boxed{2.43\,\text{k}\Omega}. \end{aligned}$$

With $U = 4$ unit capacitors,

$$\begin{aligned} C_{\text{total}} &= A_{\text{top}} C_{\text{density}} U \\ &= (10\,\mu\text{m})(10\,\mu\text{m})\left(1\,\frac{\text{fF}}{\mu\text{m}^2}\right) 4 \\ &= \boxed{400\,\text{fF}}. \end{aligned}$$



$$\begin{aligned} \frac{V_o}{V_i} &= \frac{1}{1+sRC} = \frac{1}{1+s(9.72\times10^{-10}\,\text{s})}, \\ A_{\text{DC}} &= 1 = 0\,\text{dB}, \\ \omega_p &= -\frac{1}{RC} \simeq -1.03\times10^9\,\text{rad/s}, \\ f_p &= \frac{1}{2\pi RC} \simeq \boxed{164\,\text{MHz}}. \end{aligned}$$



One pole; no finite zero

Scratch Paper

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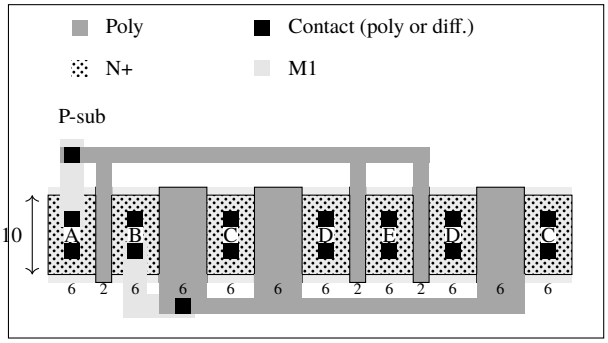
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_A = 2\,\text{V}$, $V_B = 1.2\,\text{V}$, $V_C = 0\,\text{V}$, $V_D = 1.2\,\text{V}$, $V_E = 2\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}). The dimensions are given in μm, with the poly gates having an $L_D = 0.1\,\mu$m.

a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors.

b) What region(s) are the transistors operating in?

c) **For node B only**, give an expression and calculate the total gate cap.

d) **For node D only**, give an expression and calculate the total junction cap, including the relevant channel–bulk cap. Note for the perimeter terms, include the sides underneath the gate.



(a) Parallel combinations. The six consecutive devices along the strip are

terminals	A:B	B:C	C:D	D:E	E:D	D:C
W/L	10/2	10/6	10/6	10/2	10/2	10/6

Combining the parallel C–D and D–E devices gives the circuit at right. All W/L values are drawn dimensions in μm/μm.

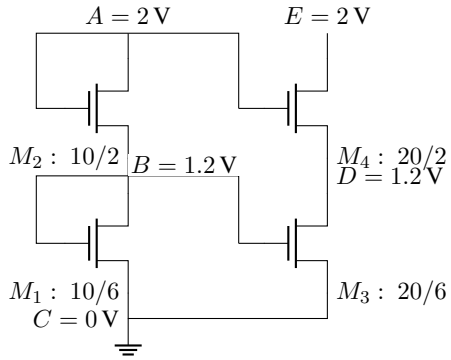
(b) Operating regions. For M_1,

$$V_{GS} = V_{DS} = 1.2\,\text{V} > V_T, \quad V_{DS} > V_{GS} - V_T.$$

Hence M_1 is in saturation. M_3 has the same voltages and is also in saturation. For M_2,

$$V_{GS} = V_{DS} = 0.8\,\text{V} > V_T, \quad V_{DS} > V_{GS} - V_T.$$

Thus M_2 and, similarly, M_4 are in saturation.



(c) Gate capacitance at B. For M_1, its gate–drain overlap is shorted out because both terminals are at B. With $L_{\text{eff}} = 6 - 2L_D = 5.8\,\mu$m,

$$\begin{aligned} C_{gB,M_1} &= \frac{2}{3}(10)(5.8)\,\mu\text{m}^2 C_{ox} + 10\,\mu\text{m} C_{ov} \\ &\simeq 38.7\,\mu\text{m}^2 C_{ox} + 10\,\mu\text{m} C_{ov}. \end{aligned}$$

For M_3, both overlap capacitors contribute:

$$\begin{aligned} C_{gB,M_3} &= \frac{2}{3}(20)(5.8)\,\mu\text{m}^2 C_{ox} + 2(20)\,\mu\text{m} C_{ov} \\ &\simeq 77.3\,\mu\text{m}^2 C_{ox} + 40\,\mu\text{m} C_{ov}. \end{aligned}$$

$$\boxed{C_{gB} = 116\,\mu\text{m}^2 C_{ox} + 50\,\mu\text{m} C_{ov}.}$$

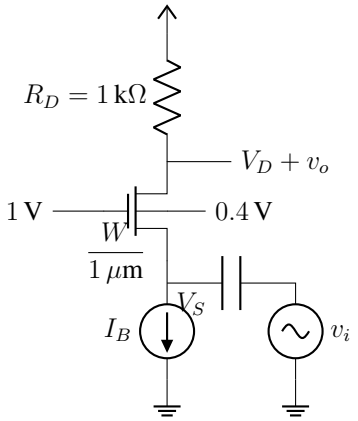
(d) Junction capacitance at D. The two D diffusion regions each have area $6\times10\,\mu\text{m}^2$ and perimeter $2(6+10)\,\mu$m, including the gate-adjacent edges. Add the channel–bulk term of M_4, with $L_{\text{eff}} = 2 - 2L_D = 1.8\,\mu$m:

$$\begin{aligned} C_{jD} &= A_D C_j + P_D C_{jsw} + C_{cb,M_4} \\ &= (6)(10)(2)\,\mu\text{m}^2 C_j + (6+10)(2)(2)\,\mu\text{m} C_{jsw} + \frac{2}{3}(20)(1.8)\,\mu\text{m}^2 C_{jc} \\ &= \boxed{120\,\mu\text{m}^2 C_j + 64\,\mu\text{m} C_{jsw} + 24\,\mu\text{m}^2 C_{jc}.} \end{aligned}$$

Problem 2 (40 points)

For the circuit shown below, the transistor is operating in a fully velocity saturated region where its DC drain current is given exactly by the given equation.

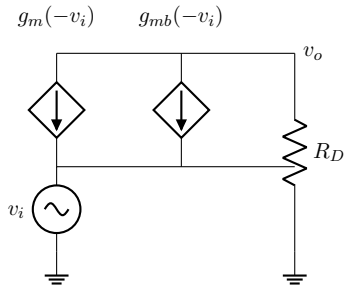
a. Draw the low-frequency small-signal model of the circuit (neglecting transistor capacitors) and give an expression for the small-signal voltage gain of the circuit, $A_v = v_o/v_i$, as a function of the relevant transistor conductances and the load resistor. Note, you can assume that the input AC-coupling capacitor acts as an ideal short.



$$I_{DS} = v_{sat}C_{ox}W(V_{GS} - V_{Tn})$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right).$$

Typeset solution. There is no V_{DS} dependence in the stated model, so $g_{ds} = 0$ and $r_o = \infty$.



$$g_m = \frac{\partial I_D}{\partial V_{GS}} = v_{sat}C_{ox}W,$$

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = v_{sat}C_{ox}W(-1)(-1)\frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} = \eta g_m,$$

$$\eta = \frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}}.$$

$$v_o = -(g_m + g_{mb})(-v_i)R_D, \qquad \boxed{A_v = \frac{v_o}{v_i} = (g_m + g_{mb})R_D.}$$

b. Assume the following DC operating points for the drain and source voltages, $V_{S,Q} = 0.2\,\text{V}$ and $V_{D,Q} = 1\,\text{V}$, and that $v_{sat}C_{ox} = 250\,\mu\text{A}/(\text{V}\cdot\mu\text{m})$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, $2\Phi_F = 0.9\,\text{V}$, and $L_D = 0.1\,\mu\text{m}$. Calculate the necessary transistor width W for $|A_v| = 5\,\text{V}/\text{V}$.

Typeset solution. Here $V_{SB} = 0.2 - 0.4 = -0.2\,\text{V}$. Consequently,

$$\eta = \frac{0.45}{2\sqrt{0.9 - 0.2}} \simeq 0.269.$$

$$A_v = (1+\eta)g_m R_D = (1+\eta)v_{sat}C_{ox}WR_D,$$

$$\boxed{W = \frac{A_v}{(1+\eta)v_{sat}C_{ox}R_D} = \frac{5}{(1.269)(250\,\mu\text{A}/(\text{V}\cdot\mu\text{m}))(1\,\text{k}\Omega)} \simeq 15.8\,\mu\text{m}.}$$

Problem 3 (20 points)

a. Poly resistor design

Design and sketch a layout of a $500\,\Omega$ poly resistor. Assume that the poly $R_\square = 60\,\Omega/\square$, each $(1\,\mu\text{m} \times 1\,\mu\text{m})$ contact has resistance of $10\,\Omega$, and fabrication tolerances limit the minimum unit resistor width $W \geq 3\,\mu$m. Use at least 2 fingers in the resistor design. In the sketch clearly label the critical dimensions.

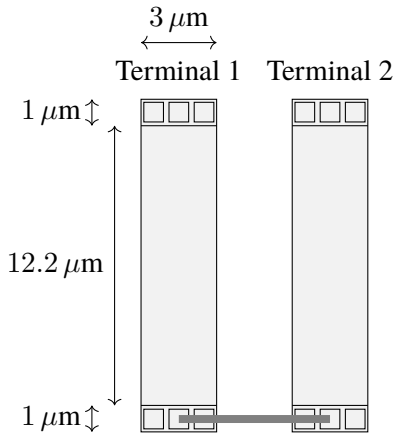
Typeset solution.

$$R_{\text{total}} = \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F, \qquad \begin{array}{l} n = \text{number of parallel contacts per end,} \\ F = \text{number of fingers (unit resistors).} \end{array}$$

$$L = \left(\frac{R_{\text{total}}}{F} - \frac{2R_c}{n}\right) \frac{W}{R_\square}.$$

Set $W = 3\,\mu$m, $n = 3$, and $F = 2$. Then

$$L = \left(\frac{500}{2} - \frac{2(10)}{3}\right)\left(\frac{3\,\mu\text{m}}{60}\right) \simeq \boxed{12.2\,\mu\text{m}}.$$



The source rounds the calculated body length $12.1667\,\mu$m to $12.2\,\mu$m; the capacitance below uses that rounded drawn dimension.

b. Poly resistor parasitic capacitance

The poly resistor has a parallel-plate parasitic capacitance to substrate of $0.1\,\text{fF}/\mu\text{m}^2$. What is the total parasitic capacitance of the resistor? Include the contact area in the capacitance calculation and assume that the capacitance is only due to the parallel-plate effect, i.e. neglect any fringing capacitance.

Typeset solution.

$$\text{Resistor area} = (12.2\,\mu\text{m} + 2\,\mu\text{m})(3\,\mu\text{m})(2) = 86.4\,\mu\text{m}^2.$$

$$\boxed{C_{\text{parasitic}} = (86.4\,\mu\text{m}^2)(0.1\,\text{fF}/\mu\text{m}^2) = 8.64\,\text{fF}.}$$

c. N-well resistor design

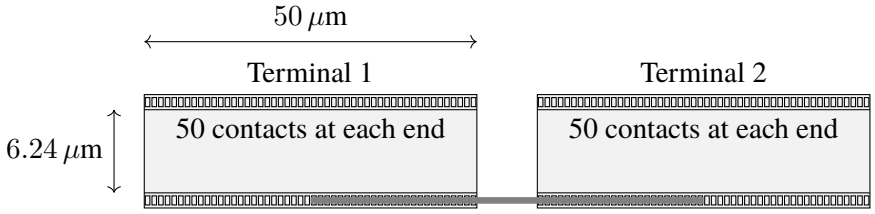
Design and sketch a layout of a $500\,\Omega$ n-well resistor. Assume that the n-well $R_\square = 2\,\text{k}\Omega/\square$, each $(1\,\mu\text{m} \times 1\,\mu\text{m})$ contact has resistance of $10\,\Omega$, and fabrication tolerances limit the minimum unit resistor length $L \geq 6\,\mu$m. Use at least 2 fingers in the resistor design. In the sketch clearly label the critical dimensions.

Typeset solution.

$$R_{\text{total}} = \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F, \qquad L = \left(\frac{R_{\text{total}}}{F} - \frac{2R_c}{n}\right) \frac{W}{R_\square}.$$

Set $F = 2$, $W = 50\,\mu$m, and $n = 50$. Thus

$$L = \left(\frac{500}{2} - \frac{2(10)}{50}\right)\left(\frac{50\,\mu\text{m}}{2000}\right) = \boxed{6.24\,\mu\text{m}}.$$



d. N-well resistor parasitic capacitance

The n-well resistor has a parallel-plate parasitic capacitance to substrate of $1\,\text{fF}/\mu\text{m}^2$. What is the total parasitic capacitance of the resistor? Include the contact area in the capacitance calculation and assume that the capacitance is only due to the parallel-plate effect, i.e. neglect any fringing capacitance.

Typeset solution.

$$\text{Resistor area} = (6.24\,\mu\text{m} + 2\,\mu\text{m})(50\,\mu\text{m})(2) = 824\,\mu\text{m}^2.$$

$$\boxed{C_{\text{parasitic}} = (824\,\mu\text{m}^2)(1\,\text{fF}/\mu\text{m}^2) = 824\,\text{fF}.}$$

e. If we have to use the $500\,\Omega$ resistor for a high-frequency application, which design should we choose, the poly or n-well design? Why?

Typeset solution. Choose the **poly resistor**: it has less parasitic capacitance.

Scratch Paper

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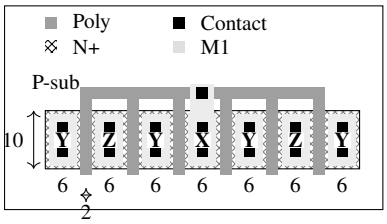
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_X = 1.2\,\text{V}$, $V_Y = 0\,\text{V}$, $V_Z = 0.4\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}). The dimensions are given in μm, with all the poly gates having a length of $2\,\mu$m and $L_D = 0.1\,\mu$m.

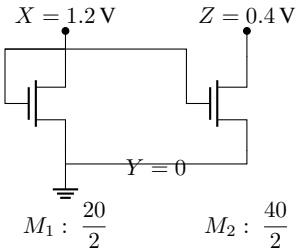
a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors. (15 points)

b) What region(s) are the transistors operating in? (5 points)

c) For node **X only**, give an expression and calculate the total gate cap. (10 points)

d) For node **Z only**, give an expression and calculate the total junction cap. Note for the perimeter terms, include the sides underneath the gate. (10 points)





a. The six unit transistors each have $W/L = 10/2$. Two are connected from X to Y and four from Z to Y; all gates connect to X. Combining them gives the two-transistor circuit above.

b. $M_1:\ V_{GS} = V_{DS} = 1.2\,\text{V} > V_T$, and $V_{DS} > V_{GS} - V_T$: saturation.

$M_2:\ V_{GS} = 1.2\,\text{V}$, $V_{DS} = 0.4\,\text{V} < 1.2 - 0.7 = 0.5\,\text{V}$: triode.

c. With $L_{\text{eff}} = 2 - 2(0.1) = 1.8\,\mu$m, M_1's gate–drain capacitance is shorted out because gate and drain both connect to X. Thus

$$C_{gX,1} = C_{gs,1} = \frac{2}{3}(20\,\mu\text{m})(1.8\,\mu\text{m})C_{ox} + (20\,\mu\text{m})C_{ov},$$
$$C_{gX,2} = C_{gs,2} + C_{gd,2} = 2\left[\frac{1}{2}(40\,\mu\text{m})(1.8\,\mu\text{m})C_{ox} + (40\,\mu\text{m})C_{ov}\right].$$

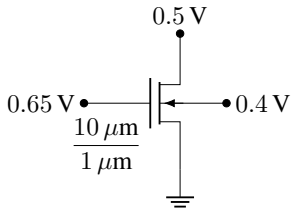
$$\boxed{C_{gX} = 96\,\mu\text{m}^2 C_{ox} + 100\,\mu\text{m}C_{ov}.}$$

d. The two Z diffusion regions have total area $A_Z = 2(6)(10) = 120\,\mu\text{m}^2$ and perimeter $P_Z = 2 \cdot 2(6+10) = 64\,\mu$m, counting the gate-adjacent sides. Following the supplied solution's capacitance notation,

$$C_{jZ} = A_Z C'_j + P_Z C'_{jsw} + C_{jc}\frac{W_2 L_{\text{eff}}}{2}$$
$$= 2(6)(10)\,\mu\text{m}^2 C'_j + 2 \cdot 2(6+10)\,\mu\text{m}C'_{jsw} + \frac{(40)(1.8)}{2}\,\mu\text{m}^2 C_{jc},$$
$$\boxed{C_{jZ}} \boxed{= 120\,\mu\text{m}^2 C'_j + 64\,\mu\text{m}C'_{jsw} + 36\,\mu\text{m}^2 C_{jc}.}$$

Problem 2 (40 points)

For the transistor below, assume $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, and $2\Phi_F = 0.9\,\text{V}$.



$$I_{DS} = I_0 \exp\left(\frac{(V_{GS}-V_{Tn})q}{\zeta kT}\right)\left(1-\exp\left(-\frac{V_{DS}q}{kT}\right)\right) \quad \text{(Subthreshold)}$$

$$I_{DS} = B\frac{W}{L-2L_D}(V_{GS}-V_{Tn})^{\alpha}(1+\lambda V_{DS})\left(2-\frac{V_{DS}}{V_{DSAT}}\right)\frac{V_{DS}}{V_{DSAT}} \quad \text{(Triode)}$$

$$I_{DS} = B\frac{W}{L-2L_D}(V_{GS}-V_{Tn})^{\alpha}(1+\lambda V_{DS}) \quad \text{(Saturation)}$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right), \qquad V_{DSAT} = V_{GS} - V_{Tn}.$$

a. Calculate V_T and state the transistor's operation region. (10 points)

$$V_T = 0.7 + 0.45\left(\sqrt{0.9-0.4} - \sqrt{0.9}\right) = \boxed{0.591\,\text{V}},$$

$$V_{GS} - V_T = 0.65 - 0.5913 = 58.7\,\text{mV}, \qquad V_{DS} = 0.5\,\text{V} > V_{GS} - V_T.$$

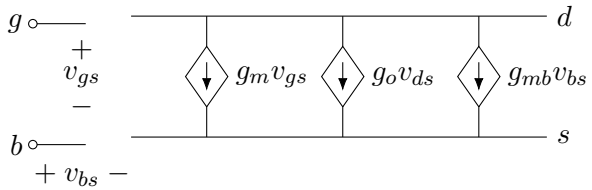
Operation region = $\boxed{\text{Saturation}}$.

b. Using the appropriate I_{DS} equation above, derive and sketch the small-signal model of the transistor. Include expressions for g_m, g_o, and g_{mb}. (20 points)

$$g_m = \left.\frac{\partial I_{DS}}{\partial V_{GS}}\right|_Q = B\frac{W}{L_{\text{eff}}}\alpha(V_{GS}-V_T)^{\alpha-1}(1+\lambda V_{DS}),$$

$$g_o = \left.\frac{\partial I_{DS}}{\partial V_{DS}}\right|_Q = B\frac{W}{L_{\text{eff}}}(V_{GS}-V_T)^{\alpha}\lambda,$$

$$g_{mb} = \left.\frac{\partial I_{DS}}{\partial V_{BS}}\right|_Q = \boxed{\frac{\gamma g_m}{2\sqrt{|2\Phi_F| + V_{SB}}}}.$$



c. Assume that $B = 65\,\mu\text{A/V}^{\alpha}$, $\lambda = 0.1\,\text{V}^{-1}$, and $L_D = 0.1\,\mu\text{m}$. Calculate the transistor's intrinsic gain (g_m/g_o) for both $\alpha = 2$ and $\alpha = 1$. (10 points)

$$\frac{g_m}{g_o} = \frac{\alpha}{V_{GS}-V_T}\frac{1+\lambda V_{DS}}{\lambda}.$$

$$\alpha = 2: \quad \frac{2}{58.7\,\text{mV}}\frac{1+(0.1)(0.5)}{0.1\,\text{V}^{-1}} \simeq \boxed{358\,\text{V/V}}, \qquad \alpha = 1: \quad \boxed{179\,\text{V/V}}.$$

Problem 3 (20 points)

Sketch a layout that matches three capacitors of unit size 2.2, 4.5, and 6.8. Assume that the unit capacitors are sized $5\,\mu\text{m} \times 5\,\mu\text{m}$. Make sure to give the non-unit capacitor(s) dimensions. In the sketch clearly label the critical dimensions and use at least 2 layout matching techniques. Also, write specifically the 2 layout matching techniques that you are using. Note, if you use a common-centroid layout technique, it doesn't have to have perfect center-of-mass matching, but it should be close.

Need to match perimeter/area for all capacitors. A unit capacitor has $A_u = 25\,\mu\text{m}^2$ and $P_u/A_u = 4/5\,\mu\text{m}^{-1}$.

$$\begin{aligned} C_1 = 2.2C_u: &\quad 1 \text{ unit cap} + 1 \text{ non-unit cap}, &\quad A_{nu,1} = 1.2A_u, \\ C_2 = 4.5C_u: &\quad 3 \text{ unit caps} + 1 \text{ non-unit cap}, &\quad A_{nu,2} = 1.5A_u, \\ C_3 = 6.8C_u: &\quad 5 \text{ unit caps} + 1 \text{ non-unit cap}, &\quad A_{nu,3} = 1.8A_u. \end{aligned}$$

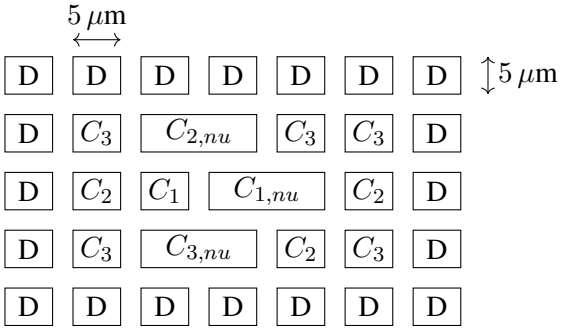
For a non-unit capacitor of area kA_u, equal perimeter-to-area ratio gives

$$x_{nu}y_{nu} = kx_u^2, \qquad \frac{2(x_{nu}+y_{nu})}{x_{nu}y_{nu}} = \frac{4}{x_u}, \qquad x_u = 5\,\mu\text{m}.$$

Thus, taking y_{nu} to be the shorter side,

$$y_{nu} = x_u\left(k - \sqrt{k^2-k}\right), \qquad x_{nu} = \frac{kx_u^2}{y_{nu}}.$$

Capacitor	k	y_{nu}	x_{nu}
C_1	1.2	$3.55\,\mu$m	$8.45\,\mu$m
C_2	1.5	$3.17\,\mu$m	$11.83\,\mu$m
C_3	1.8	$3.00\,\mu$m	$15.00\,\mu$m



The non-unit rectangles have the dimensions tabulated above; the placement sketch is not to scale. Capacitors with the same C_i label are connected in parallel. D denotes a dummy capacitor.

Layout Technique #1 = Dummies around the entire perimeter.

Layout Technique #2 = Approximately common-centroid placement.

Scratch Paper